DS ASSIGNMENT 3 -KARISHUMA VIKRAM

1. **Store n elements in an array using dynamic memory allocation and find the largest number.**

```
#include <stdio.h>
#include <stdlib.h>

int main() {
   int n, i, largest;
   int *arr;

   printf("Enter the number of elements: ");
   scanf("%d", &n);

   arr = (int *)malloc(n * sizeof(int));

   printf("Enter %d elements:\n", n);
   for (i = 0; i < n; i++) {
      scanf("%d", &arr[i]);
   }

   largest = arr[0];

   for (i = 1; i < n; i++) {
      if (arr[i] > largest)
         largest = arr[i];
   }

   printf("Largest number = %d\n", largest);

   free(arr);

   return 0;
}
```

Output

```
Enter the number of elements: 5
Enter 5 elements:
10
40
90
34
20
Largest number = 90


=== Code Execution Successful ===
```

2. **Print an array in reverse order using pointers.**

```
#include <stdio.h>

int main() {
   int arr[100], n, i;
   int *ptr;

   printf("Enter the number of elements: ");
   scanf("%d", &n);

   printf("Enter the elements:\n");
   for (i = 0; i < n; i++) {
      scanf("%d", &arr[i]);
   }

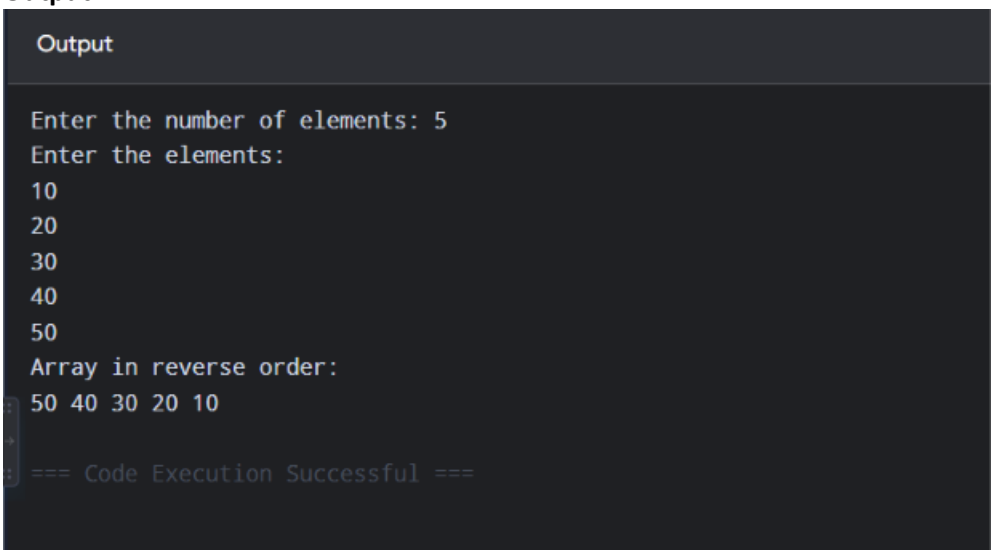
   ptr = &arr[n - 1];

   printf("Array in reverse order:\n");

   for (i = n - 1; i >= 0; i--) {
      printf("%d ", *ptr);
      ptr--;
   }

   return 0;
}
```

Output



3. **Write a C program to store information of students using structure. Sort them by name.**

```
#include <stdio.h>
#include <string.h>

struct Student {
   char name[50];
   int rollno;
   float marks;
};

int main() {
   struct Student s[100], temp;
   int n, i, j;

   printf("Enter number of students: ");
   scanf("%d", &n);

   for (i = 0; i < n; i++) {
      printf("\nEnter details of student %d:\n", i + 1);

      printf("Name: ");
      scanf(" %[^\n]", s[i].name);

      printf("Roll number: ");
      scanf("%d", &s[i].rollno);

      printf("Marks: ");
      scanf("%f", &s[i].marks);
   }

   for (i = 0; i < n - 1; i++) {
      for (j = i + 1; j < n; j++) {
         if (strcmp(s[i].name, s[j].name) > 0) {
            temp = s[i];
            s[i] = s[j];
            s[j] = temp;
         }
      }
   }

   printf("\nStudents sorted by name:\n");

   for (i = 0; i < n; i++) {
      printf("%s\t%d\t%.2f\n",
           s[i].name, s[i].rollno, s[i].marks);
   }

   return 0;
```

}

Output

```
Enter number of students: 2

Enter details of student 1:
Name: karishuma vikram
Roll number: 101
Marks: 97

Enter details of student 2:
Name: rishita vikram
Roll number: 102
Marks: 100

Students sorted by name:
karishuma vikram    101 97.00
rishita vikram  102 100.00


=== Code Execution Successful ===
```

4. **Write a C program to add n distances given in feet-inch using structure.**

```
#include <stdio.h>

struct Distance {
   int feet;
   int inch;
};

int main() {
   struct Distance d[100], sum = {0, 0};
   int n, i;

   printf("Enter number of distances: ");
   scanf("%d", &n);

   for (i = 0; i < n; i++) {
      printf("\nEnter distance %d:\n", i + 1);

      printf("Feet: ");
      scanf("%d", &d[i].feet);

      printf("Inches: ");
      scanf("%d", &d[i].inch);

      sum.feet += d[i].feet;
      sum.inch += d[i].inch;
```

```
    }

    sum.feet += sum.inch / 12;
    sum.inch = sum.inch % 12;

    printf("\nTotal distance = %d feet %d inches\n",
        sum.feet, sum.inch);

    return 0;
}
```

Output

```
Output

Enter number of students: 2

Enter details of student 1:
Name: karishuma vikram
Roll number: 101
Marks: 97

Enter details of student 2:
Name: rishita vikram
Roll number: 102
Marks: 100

Students sorted by name:
karishuma vikram    101 97.00
rishita vikram  102 100.00


=== Code Execution Successful ===
```

5. **Add two complex numbers by passing structure to a function.**

```
#include <stdio.h>

struct Complex {
    float real;
    float imag;
};

struct Complex add(struct Complex c1, struct Complex c2) {
    struct Complex result;

    result.real = c1.real + c2.real;
    result.imag = c1.imag + c2.imag;

    return result;
```

```
}

int main() {
   struct Complex c1, c2, sum;

   printf("Enter real and imaginary parts of first complex number: ");
   scanf("%f %f", &c1.real, &c1.imag);

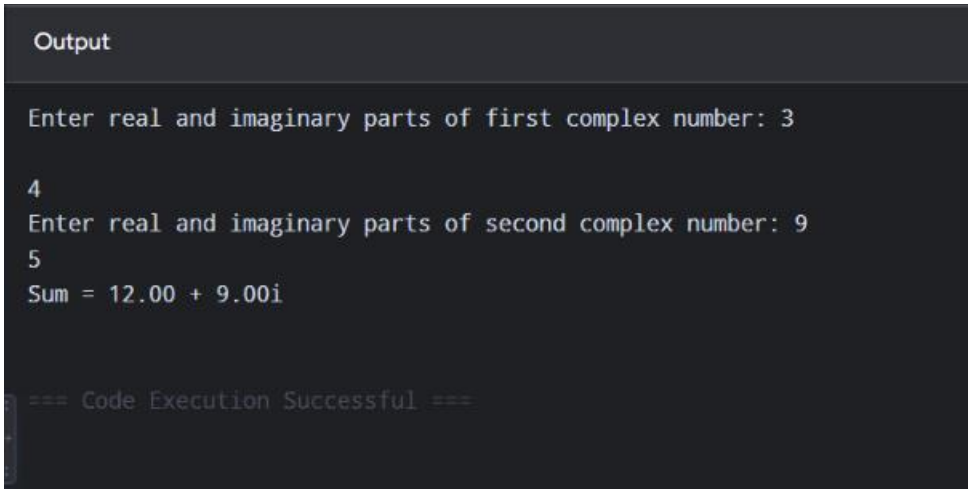
   printf("Enter real and imaginary parts of second complex number: ");
   scanf("%f %f", &c2.real, &c2.imag);

   sum = add(c1, c2);

   printf("Sum = %.2f + %.2fi\n", sum.real, sum.imag);

   return 0;
}
```

Output



6. Write a C program to maintain an Inventory Management System.

```
#include <stdio.h>

struct Product {
   int id;
   char name[50];
   int quantity;
   float price;
};

int main() {
   struct Product p[50];
   int n, i;

   printf("Enter number of products: ");
   scanf("%d", &n);
```

```
    for (i = 0; i < n; i++) {
        printf("\nEnter details of product %d\n", i + 1);

        printf("Product ID: ");
        scanf("%d", &p[i].id);

        printf("Product Name: ");
        scanf("%s", p[i].name);

        printf("Quantity: ");
        scanf("%d", &p[i].quantity);

        printf("Price: ");
        scanf("%f", &p[i].price);
    }

    printf("\n--- Inventory Details ---\n");
    printf("ID\tName\tQuantity\tPrice\n");

    for (i = 0; i < n; i++) {
        printf("%d\t%s\t%d\t\t%.2f\n",
               p[i].id, p[i].name,
               p[i].quantity, p[i].price);
    }

    return 0;
}
```

Output

```
Output

Enter number of products: 2

Enter details of product 1
Product ID: 101
Product Name: pencil
Quantity: 3
Price: 5

Enter details of product 2
Product ID: 102
Product Name: eraser
Quantity: 5
Price: 9

--- Inventory Details ---
ID  Name    Quantity    Price
101 pencil  3       5.00
102 eraser  5       9.00


=== Code Execution Successful ===
```

7. **Write a C program for decimal to binary, octal and hexadecimal conversion.**

```
#include <stdio.h>

int main() {
   int n, i, rem;
   int binary[32], octal[32];

   printf("Enter a decimal number: ");
   scanf("%d", &n);

   printf("Binary: ");

   if (n == 0)
      printf("0");
   else {
      int temp = n, count = 0;

      while (temp > 0) {
         binary[count++] = temp % 2;
         temp = temp / 2;
      }

      for (i = count - 1; i >= 0; i--)
```

```
            printf("%d", binary[i]);
        }

        printf("\nOctal: %o", n);
        printf("\nHexadecimal: %X", n);

        return 0;
    }
```

Output

```
Output

Enter a decimal number: 25
Binary: 11001
Octal: 31
Hexadecimal: 19

=== Code Execution Successful ===
```

8. **Write a C program to calculate the difference between two time periods.**

```
#include <stdio.h>

struct Date {
   int day;
   int month;
   int year;
};

int daysInMonth(int month, int year) {
   if (month == 2) {
      if ((year % 400 == 0) ||
         (year % 4 == 0 && year % 100 != 0))
         return 29;
      else
         return 28;
   }

   if (month == 4 || month == 6 ||
```

```
        month == 9 || month == 11)
        return 30;

    return 31;
}

long int dateToDays(struct Date d) {
    long int days = d.day;

    for (int y = 1; y < d.year; y++) {
        if ((y % 400 == 0) ||
            (y % 4 == 0 && y % 100 != 0))
            days += 366;
        else
            days += 365;
    }

    for (int m = 1; m < d.month; m++)
        days += daysInMonth(m, d.year);

    return days;
}

int main() {
    struct Date d1, d2;
    long int diff;

    printf("Enter first date (dd mm yyyy): ");
    scanf("%d %d %d", &d1.day, &d1.month, &d1.year);

    printf("Enter second date (dd mm yyyy): ");
```

```
    scanf("%d %d %d", &d2.day, &d2.month, &d2.year);

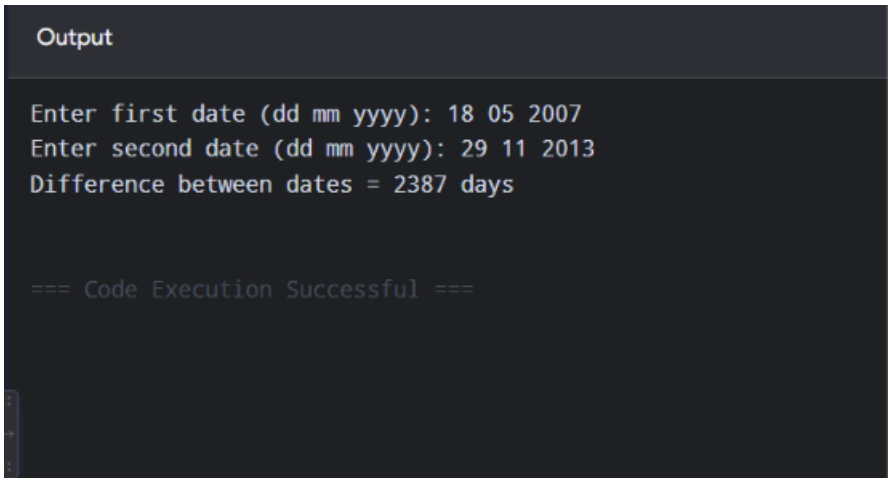
    diff = dateToDays(d1) - dateToDays(d2);

    if (diff < 0)
        diff = -diff;

    printf("Difference between dates = %ld days\n", diff);

    return 0;
}
```

Output



9. **Write a C program to create an Employee Management System using structures.**

```
#include <stdio.h>

struct Employee {
    int id;
    char name[50];
    float salary;
};

int main() {
    struct Employee emp[50];
    int n, i;

    printf("Enter number of employees: ");
    scanf("%d", &n);
```

```
    for (i = 0; i < n; i++) {
        printf("\nEnter details of Employee %d\n", i + 1);

        printf("Employee ID: ");
        scanf("%d", &emp[i].id);

        printf("Employee Name: ");
        scanf("%s", emp[i].name);

        printf("Salary: ");
        scanf("%f", &emp[i].salary);
    }

    printf("\n--- Employee Details ---\n");
    printf("ID\tName\tSalary\n");

    for (i = 0; i < n; i++) {
        printf("%d\t%s\t%.2f\n",
               emp[i].id,
               emp[i].name,
               emp[i].salary);
    }

    return 0;
}
```

Output

```
Output

Enter number of employees: 2

Enter details of Employee 1
Employee ID: 101
Employee Name: tom
Salary: 20000

Enter details of Employee 2
Employee ID: 102
Employee Name: harry
Salary: 30000

--- Employee Details ---
ID  Name    Salary
101 tom 20000.00
102 harry   30000.00


=== Code Execution Successful ===
```

10. **Write a C program to calculate area of rectangle and triangle using pointers**

```
#include <stdio.h>

float rectangle(float length, float breadth) {
   return length * breadth;
}

float triangle(float base, float height) {
   return 0.5 * base * height;
}

int main() {
   float a, b, area;

   // Function pointer
   float (*ptr)(float, float);

   printf("Enter length and breadth of rectangle: ");
   scanf("%f %f", &a, &b);

   ptr = rectangle;
   area = ptr(a, b);

   printf("Area of Rectangle = %.2f\n", area);
```

```
    printf("\nEnter base and height of triangle: ");
    scanf("%f %f", &a, &b);

    ptr = triangle;
    area = ptr(a, b);

    printf("Area of Triangle = %.2f\n", area);

    return 0;
}
```

Output

